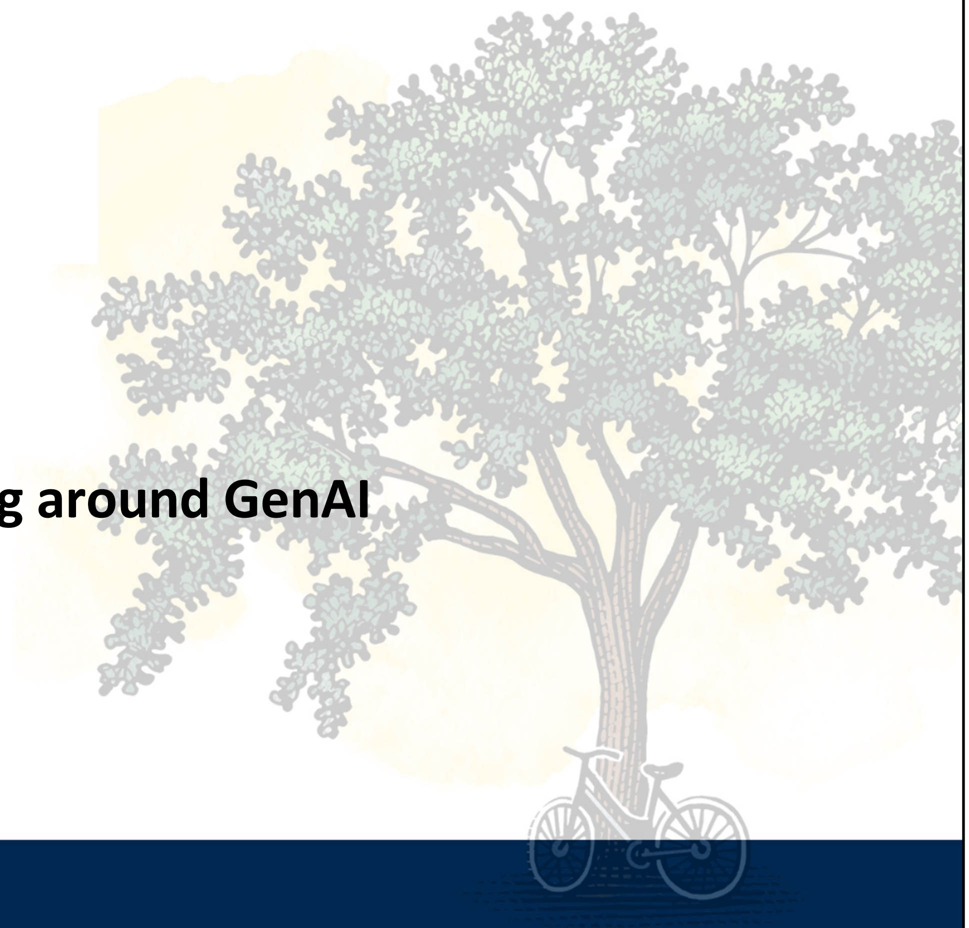
UC DAVIS
CENTER FOR EDUCATIONAL EFFECTIVENESS
Office of Undergraduate Education

Context, Metacognition and Cues:

Engaging Students in Critical Thinking around GenAI



Patricia Turner, PhD | pturner@ucdavis.edu
September 2024

About me

Patricia Turner, Ph.D.

Home • People • Patricia Turner, Ph.D.

Education Specialist

(530) 752-5173 | pturner@ucdavis.edu

The Grove, 1342

Patricia Turner, Ph.D., is an Education Specialist with the Center for Educational Effectiveness (CEE). Patricia earned her Ph.D. in Applied Linguistics from UCLA, where her areas of concentration were Conversation Analysis and institutional talk-in-interaction. Patricia's MA is in Linguistics, with emphases in teaching English as a Second Language and writing instruction.

Patricia has 26 years of teaching experience and has taught at all three systems of public higher education in California. She has held faculty positions at UC San Diego, San Diego State University, Santa Monica College and she taught extensively at UCLA while completing her doctoral studies. She has also taught at Harvard's Institute for English Language Programs, and prior to coming to UC Davis, she provided professional development for K-12 educators at WestEd, a non-profit research, development and service agency. She is interested in how modalities, dispositions, practices and artifacts of teaching and assessment influence teaching effectiveness in higher education.

At CEE, Patricia enjoys collaborating with faculty, graduate students and staff on the design and implementation of engaging, student-centered instruction with the goal of providing educational experiences that result in deep learning for diverse groups of students. Patricia received her B.A. in Psychology at UC Davis and is happy to return to UCD to engage in work on learning and teaching.

Recent Peer-Reviewed Scholarly Activity

Turner, P. (2022, March 14). Revisiting Camera Use in Live Remote Teaching: Considerations for Learning and Equity. *EDUCAUSE Review*.

Turner P. and Merrill, M. (2021, May 25). Using Structure to Promote Equity and Engagement in Live Remote Sessions. *EDUCAUSE Review*.

Turner, P. and Rossi, M. (2021). Promoting Students' Engagement with Disciplinary Texts as Inclusive Teaching Practice. *IDEA Papers*. IDEA Paper 85.

AI news

ARTIFICIAL INTELLIGENCE

OpenAI has released a new ChatGPT bot that you can talk to

The voice-enabled chatbot will be available to a small group of people today, and to all ChatGPT Plus users in the fall.

By Melissa Heikkilä July 30, 2024

[MIT Technology Review, July 30, 2024]

OpenAI says Iranian group used ChatGPT to try to influence U.S. election

The AI company found its chatbot was used to make content for websites and social media posts aimed at increasing U.S. political polarization but says the effort didn't gain traction.

[Washington Post, August 16, 2024]

Business / Tech

OpenAI worries people may become emotionally reliant on its new ChatGPT voice mode

By Clare Duffy, CNN
3 minute read · Published 5:33 PM EDT, Thu August 8, 2024

[CNN, August 8, 2024]

> During early testing, including red teaming and internal user testing, we observed users using language that might indicate forming connections with the model. For example, this includes language expressing shared bonds, such as "This is our last day together." While these instances appear benign, they signal a need for continued investigation into how these effects might manifest over longer periods of time. More diverse user populations, with more varied needs and desires from the model, in addition to independent academic and internal studies will help us more concretely define this risk area.

Open AI: [ChatGPT 4o System Card, Aug. 8, 2024]

AI news

DIGITAL DEMOCRACY
CALMATTERS

Search for bills, hearings, people, and more

Issues Directories Legislators Data Sources & Methodology

Bills

SB 1047: Safe and Secure Innovation for Frontier Artificial Intelligence Models Act.

Session Year: 2023-2024 **House:** Senate

CalMatters Digital Democracy, Accessed Aug 14, 2024

Artificial Intelligence

OpenAI whistleblowers ask SEC to investigate alleged restrictive non-disclosure agreements

By **Chandni Shah**

July 15, 2024 7:27 AM PDT · Updated a month ago

Aa

Reuters, July 15, 2024

Critical Thinking (Mason 2007)

Skills, Dispositions & Knowledge

Skills of *critical reasoning* (e.g., inferring, evaluating, reasoning)

Dispositions
A *critical attitude* (scepticism, tendency to ask **probing questions**)

A *moral orientation* that motivates critical thinking

Knowledge of particular content
Concepts in critical thinking (principles in logical reasoning)

Disciplinary knowledge

Overcoming egocentric/ sociocentric thinking (Paul, 1982)

Ability to **think critically about one's own positions**, arguments, assumptions, and world view
➜ **deep knowledge of oneself**

Ability to think critically about positions other than one's own

Integrated critical thinking

"Thinking that is **not entrenched in dogma** (although committed to reason), is **willing to consider multiple perspectives**, is **informed**, **sceptical**, and **entails sound reasoning**"
(Mason, 2007, p. 344).

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Critical Thinking (Mason 2007)

Skills, Dispositions & Knowledge

Cues

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Self-regulated learning

“[S]tudents’ **active control** of their cognitive, motivational, and behavioral engagement in learning” (Wolters and Brady, 2021:1321).

Involves **awareness of thinking**, **use of strategies**, and **sustained motivation** (Paris and Winograd, 2003).

“...[A] **positive** set of attitudes, strategies , and motivations for enhancing thoughtful engagement with tasks” (Paris and Winograd, 2003:5).

Goal: Self-regulation with regard to use of Generative AI, to encourage **thoughtful use**, **use with integrity**, and **use that advances student learning**.

Context: Module on AI

Instructional Sequence

23 graduate students and postdocs experienced a pedagogical module on the use of GenAI in education.

- Pre-module survey
- Four videos on GenAI
- 75-minute interactive lecture
- Peer Discussion
- **Cognitive Process Analysis based on Bloom's Revised Taxonomy**
- Discussion post
- Post-module survey

VIDEO 1

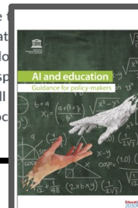
Ethan Mollick (Professor of Management, the Wharton School) and Lilach Mo[...] become well-known for their work on Artificial Intelligence in Education. They [...] and students. We'll watch the first two videos. The rest can be found on YouTu[...] Business School."

WATCH: Part 1: Practical AI for Instructors and Students (10:17 min listen) (Wharton School of the University of Pennsylvania)

this week.

Benefits and Risks of using GenAI in Education

Benefits (Mollick and Mollick, 2023)	Challenges/Risks (Bailey, 2023)
Mentor - Providing feedback	Student cheating
Tutor - Providing direct instruction	Biases in algorithms
Coach - Supporting reflection and metacognition	Privacy concerns
Teammate - Providing alternate viewpoints	Decreased social connection
Student - Receiving clarifications and teaching others	Overreliance on technology
Simulator - Offering practice and application	Equity issues
Tool - Accomplishing lower level tasks	



On Function
"We also need to question what the introduction of AI into education might achieve: What are the real benefits AI might bring? How do we ensure that AI meets real needs, and is not just the latest EdTech fad? What should we allow AI to do?" (p. 13).

On Exacerbating Inequalities
"[I]t will be increasingly important for every citizen to have the opportunity to develop a robust understanding of AI - what it is, how it works and how it might impact on their lives. This is sometimes called "AI Literacy." For this, teachers will play a key role...." (p. 23).

On Evidence
"In fact, the purported efficacy of many AI tools may be due more to their novelty than their substance. We simply do not have sufficient evidence" (p. 26, citing Holmes, et al. 2018).

(UNESCO, 2023)

Exploring ChatGPT

Step 3

Analysis

Activity	What cognitive processes did the activity engage you in when you did this without the help of ChatGPT? (Method 1)	What cognitive processes did the AI take on for you when you used AI? (Method 2)	Which activity do you feel was more effective at advancing your learning/retention/mastery?	Which method would you choose for your students' learning if they had to do a similar activity in your class?	Other thoughts?
Scenario 1: You have been asked to give a short presentation on one of the three major learning theories (behaviorism, cognitivism and constructivism) to the TAs in your department.					
Scenario 2: You have been asked to write a short explanation of the Apprenticeship of Observation and give your thoughts on the appropriateness of instructors' using the Apprenticeship of Observation as a basis for one's teaching decisions?					

[Center for Humane Technology] KEY ISSUES SOLUTIONS

Together we can align technology with humanity's best interests.

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Context: Module on AI


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and students. We'll watch the first two videos. The rest can be found on YouTu
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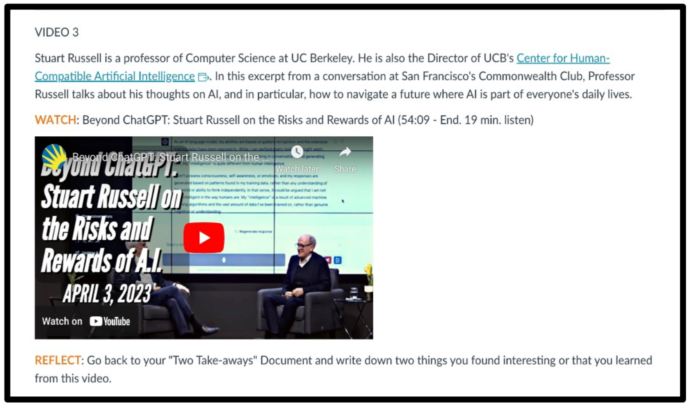
WATCH: Part 1: Practical AI for Instructors and Students (10:17 min listen) (Wharton School of the University of Pennsylvania)

REFLECT: What are two take-aways from this video -- two things you learned from watching this video? Click on this link to create a "Two Take-Aways" document that you can use for the rest of this unit to record to your responses to the videos. You will be prompted to make a copy that will be private to you and live in your Google Drive. You may find this document helpful if you decide to write a Commentary this week.

Introduction to AI: Mollick & Mollick,
Wharton School of Business


VIDEO 3

Stuart Russell is a professor of Computer Science at UC Berkeley. He is also the Director of UCB's Center for Human-Compatible Artificial Intelligence. In this excerpt from a conversation at San Francisco's Commonwealth Club, Professor Russell talks about his thoughts on AI, and in particular, how to navigate a future where AI is part of everyone's daily lives.

WATCH: Beyond ChatGPT: Stuart Russell on the Risks and Rewards of AI (54:09 - End. 19 min. listen)

REFLECT: Go back to your "Two Take-aways" Document and write down two things you found interesting or that you learned from this video.

Beyond ChatGPT: Stuart Russell, UC Berkeley
Center for Human-compatible Artificial Intelligence

The AI Dilemma: Navigating the road ahead with Tris
Center for Humane Technology

AI Use

AI Ethics/
Safety

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Cognitive Process Analysis: The Task

11 students engaged with a task that led them through **an analysis of their own experiences as a learner using ChatGPT**.

- **Method 1**: Worked on a task first alone, then in groups, using available resources (e.g., Canvas, Internet)
- **Method 2**: Worked on the same task using ChatGPT

http://tinyurl.com/chatgptexview

First, **without using** ChatGPT

Method 1

Activity-Groups 1 & 2	Please begin drafting your presentation here. You can consult notes, our lecture slides on Canvas, colleagues, and/or the Internet.
Scenario 1: You have been asked to give a short presentation on one of the three major learning theories you learned about in Week 1 (behaviorism, cognitivism and constructivism) to the TAs in your department.	

Then, **using** ChatGPT

Method 2

Activity-Groups 1 & 2	Ask ChatGPT for help with this assignment and paste its response here. Feel free to edit.
Scenario 1: You have been asked to give a short presentation on one of the three major learning theories you learned about in Week 1 (behaviorism, cognitivism and constructivism) to the TAs in your department.	

Cognitive Process Analysis

What cognitive processes did the activity engage you in when you worked without ChatGPT?

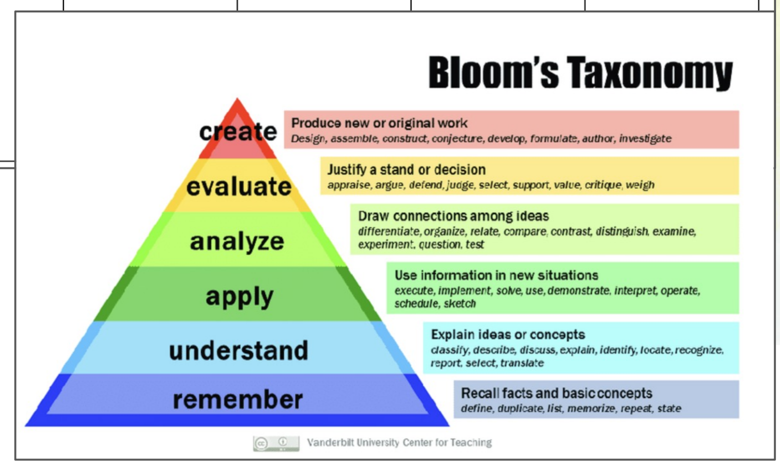
What cognitive processes did the AI take on for you when you used ChatGPT?

Which activity do you feel was more effective at advancing your learning/retention/mastery?

Which method would you choose for your students' learning if they had to do a similar activity in your class?

Analysis

Activity	What cognitive processes did the activity engage you in when you did this without the help of ChatGPT? (Method 1)	What cognitive processes did the AI take on for you when you used AI? (Method 2)	Which activity do you feel was more effective at advancing your learning/retention/ mastery?	Which method would you choose for your students' learning if they had to do a similar activity in your class?	Other thoughts?
Scenario 1: You have been asked to give a short presentation on one of the three major learning theories (behaviorism, cognitivism and constructivism) to the TAs in your department.					



Cues: Critical Thinking Questions

12 Replies

Week 8 | Commentary: Generative Artificial Intelligence in Education

To wrap up this module, take some time to reflect on what we discussed in class and to synthesize what you have learned in the module(s) thus far. Note that only 5 commentaries are required to pass the course and/or receive a certificate.

Discussion Board

This week, we have explored and discussed information and recommendations on Generative AI from several perspectives in an attempt to gain a sense of the current conversation around GenAI and how it can inform our thoughts on GenAI in education. For this week's commentary, please consider the following question:

Taking into account the videos in this module, the information presented on the class slides, and the analysis that we did in class of the cognitive and academic processes you engaged in *with* and *without* ChatGPT's assistance (if you were able to be here for that - if not, no problem; you can still do do this commentary without that), and any other knowledge you may have, **what is your current thinking about how we, as educators, should approach the use of Generative AI in the university classroom? Should we use it? If so, how? What should instructors take into account when using GenAI in their teaching?** Feel free to review the materials to make your response more specific. You might also want to take a look at your "My Take-aways" document that you created during the module.

To receive full credit, please:

- Respond to the question in a 100-150 word text.
- Incorporate another student's answer into your response (e.g., "Like Lee, I think that...") OR respond to another student's post (e.g., Lee - Thanks for that idea about X....or Hi Lee, I have questions about X, too...etc.)

Please remember to observe our netiquette guidelines.

Reply

What is your current thinking about how we, as educators, should approach the use of Generative AI in the university classroom? Should we use it? If so, how? What should instructors take into account when using GenAI in their teaching?

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Cues: Critical Thinking/Analytical Questions

Analyzing Perspectives on Generative AI

	Wharton School of Business	**Center for Human-compatible Technology (Stuart Russell, UCB)**	**Center for Humane Technology**
What **values** do you see embedded in each of the approaches to artificial intelligence (Wharton School, Center for Human-compatible Technology, and Center for Humane Technology?			
How do you know that the speakers hold these values (based on what they say)?			
What **view** of artificial intelligence does each seem to take? What **specific arguments** do they make to support their view?			
What **assumptions** is each making about artificial intelligence? How do these assumptions shape their arguments?			
What are some **potential problems** with each perspective?			

Next steps

- What **values** do you see embedded in each of the approaches to artificial intelligence
  - How do you know that the speakers hold these values (based on what they say)?
- What **view** of artificial intelligence does each seem to take?
  - What **specific arguments** do they make to support their view?
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  - How do these assumptions shape their arguments?
- What are some **potential problems** with each perspective?

Goal: Self-regulation with regard to use of Generative AI, to encourage **thoughtful use**, **use with integrity**, and **use that advances their learning**.

Student response

Nov 28, 2023 4:54pm

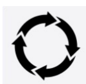
Just this week, Inside Higher Ed published a report where they found that American students and faculty rank among the lowest in the world for AI use. There is also a higher degree of suspicion regarding AI ethics and the potential for plagiarism among American university leaders than in any other country. This tells me that we have no idea how to effectively incorporate AI into the classroom, but as other students have said, we can't ignore it because it's here to stay and will likely become part of everyday life (much like the Internet). In fact, it will pervade every aspect of life whether we like it or not, and that process has already begun. Thus, I think it's worth spending time to figure out how to use it in the classroom; we should approach it as a tool that can add to our efforts, or as suggested, make tedious tasks less time consuming like a dishwasher does. My view is it should be a tool that enhances the cognitive process, and not as a replacement for thinking. For example, the activity we did in class where we did the thinking first, then added ChatGPT's response to see what we may have missed is fine; I don't think we should have started with ChatGPT for a recall activity. If we start with ChatGPT as an initial research step (like we often do with Wikipedia), we need to find ways to get students to engage with the content itself, as generative AI just gives you an answer and doesn't allow you to engage for further learning (unless you ask appropriate follow up questions of course, but it's still doing the thinking for you). Finally, I agree with about helping students become familiar with AI and adding it to their toolbox. Generative AI has taken over in the last year, so there's no doubt that students will need to use/engage with it beyond the classroom, so we should equip them with at least basic knowledge of how to do that in a way that helps them.

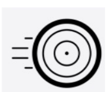
Reply | Mark as Unread

Response used with permission.

- Cites and provides a link to another **source** (Inside Higher Ed article) on GenAI
- Expresses concern about Higher Ed's ability to teach GenAI
- Notes GenAI should **enhance the cognitive process, not replace it**
- Cautions that GenAI can be "doing the thinking for you"
- Instructors should teach students about GenAI for use beyond the classroom

What next?

 Iterative process

 Moving target

 Empathy and patience with process (for students and myself!)

Images: The Noun Project
Iterative process: Aaron K. Kim
Moving Target: David Christensen, the Noun Project
Zen: Icons producer
Photo: Karin Higgins/UC Davis

Coda: 3 questions I ask myself about GenAI use

In this use case:

- Does using Generative AI diminish me **as a thinker**?
- Does using Generative AI diminish **my authenticity**?
- Does using Generative AI diminish **my relationships**?

What I’ve noticed:

- Questions are relevant each time I consider using GenAI
- Answers seem to be highly subjective, only I can answer these for myself
- Answers may change for every use case and may change over time

Reflective Tools / Resources

Cognitive Process Analysis:
http://tinyurl.com/chatgptexview

Analyzing Perspectives on GenAI - Matrix
https://tinyurl.com/AI-CTMatrix

Canvas site with GenAI module
Email me at pturner@ucdavis.edu for access

Photo by Anastasiya Romanova on Unsplash

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Useful resources

Perspectives from AI Ethicists | Foundations of Humane Technology (course - Center for Humane Technology)

Three Principles for Creating Safer AI (Center for Human-Compatible Artificial Intelligence)

Reports from National and International Organizations

Guidance for generative AI in education & research (UNESCO)

AI and education: Guidance for policy makers (UNESCO)

Beijing Consensus on Artificial Intelligence in Education (UNESCO)

Artificial Intelligence and the Future of Teaching and Research (US Department of Education)

Center for Educational Effectiveness (CEE) Resources

CEE's Generative AI Student Survey

AI and Student Writing (CEE Just-in-Time-Teaching Resource)

Introduction to Generative AI (CEE Just-in-Time-Teaching Resource)

References

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Wolters, C. A., & Brady, A. C. (2021). College students' time management: A self-regulated learning perspective. *Educational Psychology Review*, *33*(4), 1319-1351.

Thank you!

Questions or comments:
pturner@ucdavis.edu

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Students at University of California, Davis.
Credit: Elena Zukova, University of California